

COMPUTER MEMORY

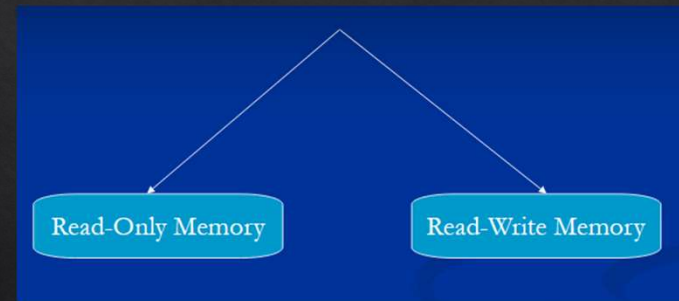
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CLASSIFICATION OF MEMORY BY PRIORITY

- a) **Primary** – storage of immediate data necessary to run the computer (RAM, ROM, and Cache)
- b) **Secondary** – long-term storage of data (HDD, SSD, Floppy Disk, etc.)
- c) **Tertiary** – disconnected storages (CD, DVD, memory card, flashdrive, external HDD, etc.)

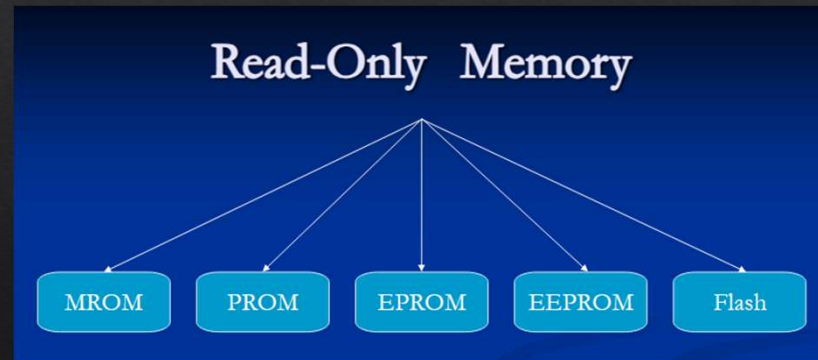
CLASSIFICATION OF MEMORY ACCESS



VOLATILITY

- a) **Non-volatile** – stores the information even without power supply
- b) **Volatile** – requires constant power to maintain the stored information

DIVISION OF READ-ONLY MEMORY



READ-ONLY MEMORY TYPES

- MROM (Mask Read-Only Memory)
- its content is programmed by manufacturer not by user
- then we can use it only for reading
- the oldest type of ROM



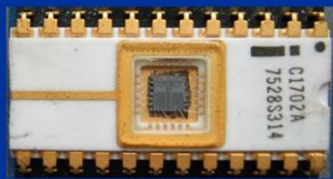
READ-ONLY MEMORY TYPES

- PROM (Programmable Read-Only Memory)
- it is blank from the manufacturer
- can be programmed just once by the user



READ-ONLY MEMORY TYPES

- EPROM (Erasable Programmable Read-Only Memory)
- can be erased by UV light
- recognizable by the transparent window in the top



READ-ONLY MEMORY TYPES

- EEPROM (Electrically Erasable Programmable Read-Only Memory)
- can be programmed and erased electrically
- while erasing whole content is erased



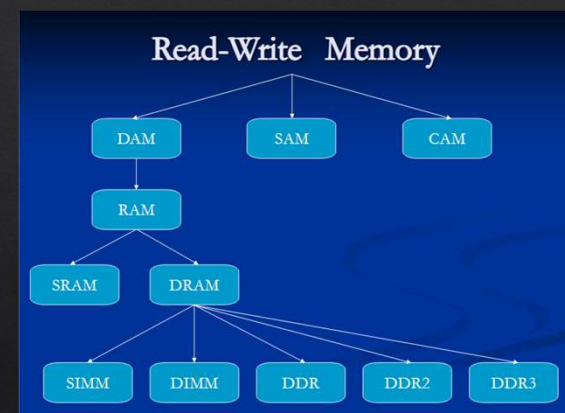
READ-ONLY MEMORY TYPES

Flash memory

- used in memory cards or USB flash drives
- specific type of EEPROM
- erased and programmed in larger blocks
- faster than old EEPROM



DIVISION OF READ-WRITE MEMORY



READ-WRITE MEMORY TYPES

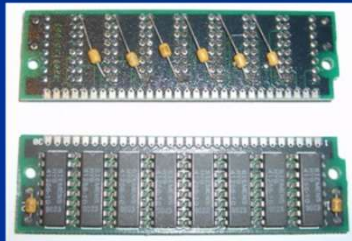
- DAM (Direct Access Memory)
 - direct access to all memory cells
 - RAM
- SAM (Sequential Access Memory)
 - data reading in sequence
 - HDD, optical discs
- CAM (Content Addressable Memory)
 - associative memory
 - returns value according to supplied word

READ-WRITE MEMORY TYPES

- RAM (Random Access Memory)
 - any piece of data can be returned in a constant time
- SRAM (Static Random Access Memory)
 - does not need to be periodically refreshed
 - more complex internal structure, more expensive, less capacity than DRAM
 - DRAM (Dynamic Random Access Memory)
 - to keep information must be periodically refreshed
 - structural simplicity, high capacity

READ-WRITE MEMORY TYPES

- SIMM (Single In-line Memory Module)
- is a type of memory module from the early 1980s to the late 1990s



READ-WRITE MEMORY TYPES

- DIMM (Dual In-line Memory Module)
- designed for use in personal computers, workstations and servers
- compatible with 64-bit Intel Pentium



READ-WRITE MEMORY TYPES

DDR

- DDR (Double Data Rate)
- used in personal computer, notebooks
- dual-channel configuration



READ-WRITE MEMORY TYPES

DDR2

- DDR (Double Data Rate)
- supersedes DDR RAM which are not compatible with DDR2
- the data bus can be run at twice the speed of the memory clock



READ-WRITE MEMORY TYPES

DDR3

- DDR (Double Data Rate)
- is an improvement over its predecessor DDR2
- no backward compatibility



END